PalmSource™ 99

Latest Palm OS® Software New Handheld APIs

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Overview

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- Command Bar
- Security and Masked Records
- UI Widgets
- Notification Manager
- Other APIs

Command Bar

- When you get menuCmdBarOpenEvent, make calls to MenuCmdBarAddButton() specifying
  - Where to add the button (right or left side)
  - The bitmap ID
  - Result action (menu command, character, etc.)
  - Text (if it isn't a menu command)
- If the result is a menu command, you're done; otherwise, write code to handle the action event

Command Bar (Cont.)

- Restrict the number of buttons to 4 so the user can see the feedback text
- Only show the relevant buttons for the user's context
- You might set the bit to stop the field from adding cut/copy/paste/undo buttons
- MenuEraseStatus() will hide the command bar; this should be called more than before

Security and Masked Records

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- prefShowPrivateRecords is now 2 bits
  - A record's setting is still boolean: public or private
- SecSelectViewStatus()
  - Shows the Security form, does password checking, returns the new status
- SecVerifyPW(newLevel)
  - Does whatever is necessary to switch to the specified security level
  - Changes system preference
  - Educates user on consequences

Table Support for Masked Rows

- Tables have built-in support for masked records
  - TblSetRowMasked, TblSetColumnMasked, TblRowMasked
  - Draws the gray box and lock icon for you
  - When you get a tblSelectEvent:

```
if (TblRowMasked(tblPtr, theRow)) {
   SecVerifyPW(showPrivateRecords)
   RedrawEverything();
}
```

New UI Widgets

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- Extended Gadgets
- Color picker
- Sliders
- Graphical Buttons

Extended Gadgets

- System calls your code for taps, drawing…
- Saves you time and code, and is called at all the right times
- PalmRez changed; no constructor change necessary
- You specify a callback for a gadget
  - FrmSetGadgetHandler(formP, index, fnP)
- Callback is passed a Ptr to gadget, the cmd, and param

Extended Gadgets (Cont.)

- Gadget Callback commands:
- formGadgetDrawCmd
  - Set gadgetP->attr.visible if you do any drawing in response to this
- formGadgetEraseCmd
  - Won't be sent unless attr.visible is true
  - If you return false, OS will erase your bounding rectangle
- formGadgetHandleEventCmd
  - For taps, etc

Extended Gadgets (Cont.)

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- **frmGadgetEnterEvent**: event sent when user taps w/pen on object; unless the app handles this,it'll be sent to the gadget's handler
- **frmGadgetMiscEvent**: you can enqueue one with the gadget ptr and gadget-specific details; event ptr gets sent to handler

Color Picker

- UIPickColor(indexP, rgbP, start…)
- Two UI modes: RGB sliders and “chits”
- If you pass both an indexP and an rgbP:
  - Both will have data returned; the rgbP may be “more correct” since it won’t change the palette so there might not be an index for the chosen rgb
  - You must specify which mode to start in

Slider Controls

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- sliderCtl and feedbackSliderCtl
  - feedbackSliderCtl sends ctlRepeatEvents
- Use CtlGetValue, CtlSetValue
  - SliderControlType * can be cast to ControlType *
- CtlGetSliderValues(), CtlSetSliderValues()
- CtlNewSliderControl()
  - Use this until GUI tools have support

Graphical Buttons

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- Like a normal button, but a bitmap instead of text, with a second “selected” image possible
- Why? Two reasons:
  - The old overlaying technique is very inefficient in multi-bit modes
  - Highlighting by color inversion is ugly
- Create with CtlNewGraphicControl, or an updated Constructor
- CtlSetGraphics() to change images at runtime

Notification Manager

- For sending notifications of an event to various parties (applications, functions, the OS, hacks, etc.)
- More efficient and extensible than sending launch codes to every application
- Various usage models
  - Simple notification
  - Collaborative (set a ‘handled’ bit)
  - Collective (add info to the structure)

Notification Manager (Cont.)

- SysNotifyRegister(), SysNotifyUnregister()
  - If you register an app, it'll send a launch code
  - If you register a function, it'll call it
  - You can request a priority order (but nothing can be guaranteed)
- SysNotifyBroadcast(), SysNotifyBroadcastDeferred()
  - Deferred means "Next EvtGetEvent()"
- Send your own codes (register a creator id)

Notification Manager (Cont.)

- Current System notifications:
  - TimeChanged
  - AntennaRaised
  - DisplayChange (palette or depth changed)
  - HotSync® process starting and ending
  - Reset finished
  - Password forgotten
  - Sleep (multiple notification stages)

Notification Manager (Cont.)

- Sleep happens in several stages:
  - Request (includes reason, and deferSleep field)
    - Reasons: PowerButton, autoOff, Resumed, Unknown
  - Notify
  - EarlyWakeup (usually not used by apps)
  - LateWakeup
- Sleep notifications aren't guaranteed to occur

Menus

- Now you can change menus at runtime!
- MenuHideItem(id)
- MenuShowItem(id)
- MenuAddItem(positionID, id, cmdChar, textP)
  - Adds after the item with id positionID
- Watch for a menuOpenEvent
- Usage by OS
  - On Japanese units, can change edit menu of an application automatically

Misc. New APIs

- ResLoadConstant
- TblGetItemPtr
- FrmNewGsi
- EvtSetAutoOffTimer
- DateTemplateToAscii
- FrmCustomResponseAlert
  - Like FrmCustomAlert but includes an input field and a ptr to a callback routine of yours

The Dynamic heap

- Mostly same memory manager as 3.2 and 3.3
- Size of the dynamic heap now depends on the amount of system RAM on card 0
  - The size used to be hard-coded into a rom, and was 32, 64, 96 or 128 for various devices
- 64k on < 2 megabyte devices
- 128k on >=2 megabyte devices
- 256k on >=4 megabyte devices

Misc. API Changes

- FtrPtrNew allows allocating >64k chunks
  - FtrPtrNew allocates in the storage heap; use DmWrite
  - Standard large-object warnings: might not be enough contiguous space; performance…
- FrmHide/ShowObject work on lists
- Can now hide the “Edit Category” line
- StrToLower works on chars other than A-Z
- ScrDisplayMode renamed to WinScreenMode

SelectTime Changes

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- “All Day” button
- 2 new parameters
  - startOfDisplay: to auto-scroll hour list
  - endOfDay: time that the “day” ends, or -1
  - (startOfDay already existed, but its meaning now makes more sense)
- Old version still at old trap location for compatibility

SDK Include Files

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- Contents have moved around, been renamed
  - #include <PalmOS.h> instead of <Pilot.h>
- Better distinction between public and private structs so you know what to use
- Privates have moved out of the SDK
- New types
  - Int -> Int16, Word -> UInt16, Long -> Int32…
  - Handle -> MemHandle, Ptr -> MemPtr
- Removal of FooPtr typedefs; use Foo *
- Still compatible backwards and forwards

Type Changes (Cont.)

- Sometimes things changed from Int to UInt16
- e.g., Int DmFindResource(…)
  - Docs say it returns index, or -1 if not found
  - Now it returns a UInt16, which is binary compatible
  - Comparing against -1 will still work, but checking <0 won't any more—but saying <0 was assuming more than what the docs said
- Table row parameters now also unsigned
- Upshot: cleaning up the OS sometimes might require you to clean up your code too, and compatibility is being retained

Other Changes (Or Not)

- Various bug fixes
- Various improvements for Internationalization and Localization (and the Overlay Manager)
- Lots of changes in lowest-level software, new architecture for hardware abstraction
  - Shouldn’t affect applications
- Desktop software, web clipping were not (perceptibly) changed

Release vs. Debug roms

- Using the debug rom in Pose catches application bugs galore
- The debug rom keeps getting more checks
  - Parameter validation
  - Form object validation
- Many ErrFatalDisplay calls are now properly NonFatal, so you'll see them but not your customers
  - On debug rom, hit "Cont" to see if your app handles the error code properly
- New specific code for debugging

Debugging Form Drawing

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- FrmDrawForm
  - It ought to erase form content, but some apps rely on drawing before calling FrmDrawForm
  - Now will always erase contents on debug roms
  - If ((saveBehind && !visible) || customAlert || modal) will WinEraseRectangle on release roms
- FrmEraseForm
  - In debug roms, doesn't restore the saveBehind bits to make the app redraw "manually" to mimic low-memory
  - Still saves the bits, to use the same amount of memory as normal

More Debug rom Differences

- Lots of managers do extra validation
- Color Debug rom uses unique UI colors
- Screen background is cleared to UICaution (currently yellow) when switching apps
- FrmCopyLabel sees if new text will fit
- DmReleaseResource does validation
- Buffers and stack get set to 0xfe or such, memory leaks are reported
- Etc.

What To Do Now

- Visit the appropriate talks and labs
- Get the CD tonight
- Try your app on the new OS (debug rom!)
- Implement new features
- Give feedback
- Join the Development Resources Seeding program (to get more roms)

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